(Please write your Enrollment

MID-TERM EXAMINATION
(Course Name B.Tech. MAE/DMAM/IT) (Semester 1st)
(October, 2023) OFF LINE mode

Subject Code:BMA 110	**Subject: Engineering Mechanics**
Time : 1 ½Hours	**Maximum Marks : 30**

Note: Q. 1 is compulsory.

Q1		**(2.5*4)**	
	(a) Draw the free body diagram of the sphere supported as shown in Figure 1.		

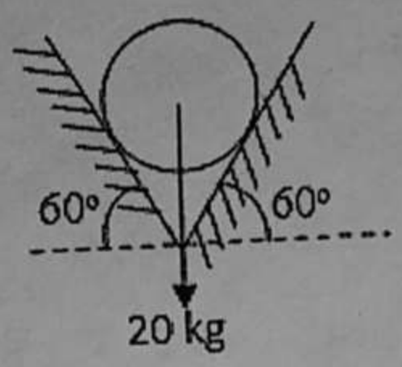


Figure 1

(b) Define angle of repose.

(c) Locate the centroid of the lamina shown in Figure 2.

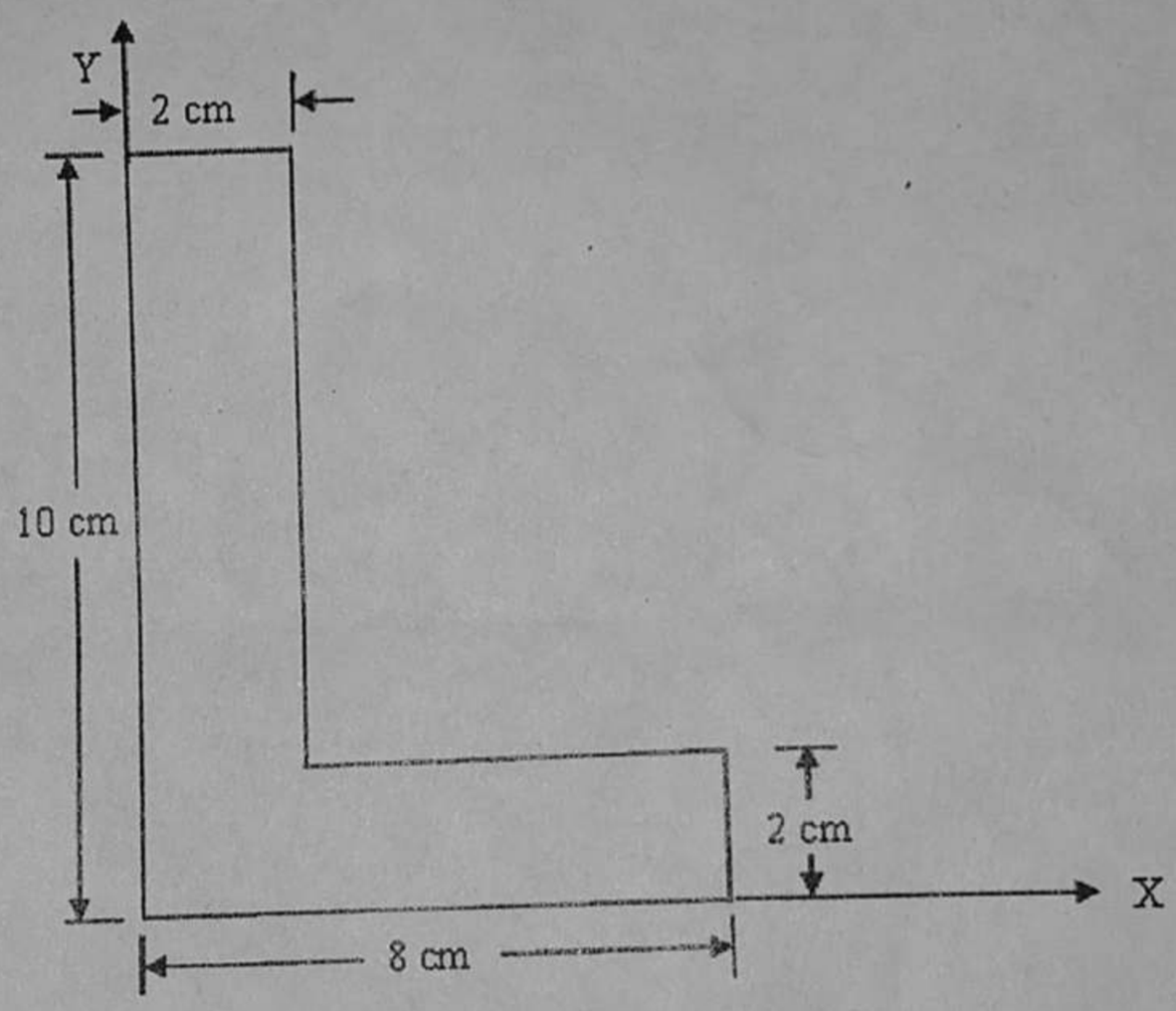


Figure 2

(d) Use proper formula to find out whether the truss shown in Figure 3 is perfect or imperfect.

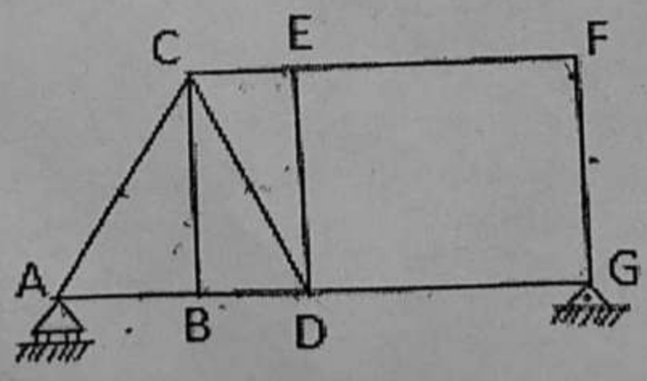


Figure 3

Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a) Two forces, P and 2P, have resultant of 260 N. If the direction of the larger force is reversed and the other remains unaltered, the resultant becomes 180 N. Determine the magnitude of the two forces and the angle between them.		
	(c) A roller of weight 800N rests on a smooth inclined plane and is kept free from rolling down by a string as shown in the Figure 4. Work out tension in the		

string and reaction at the point of contact.

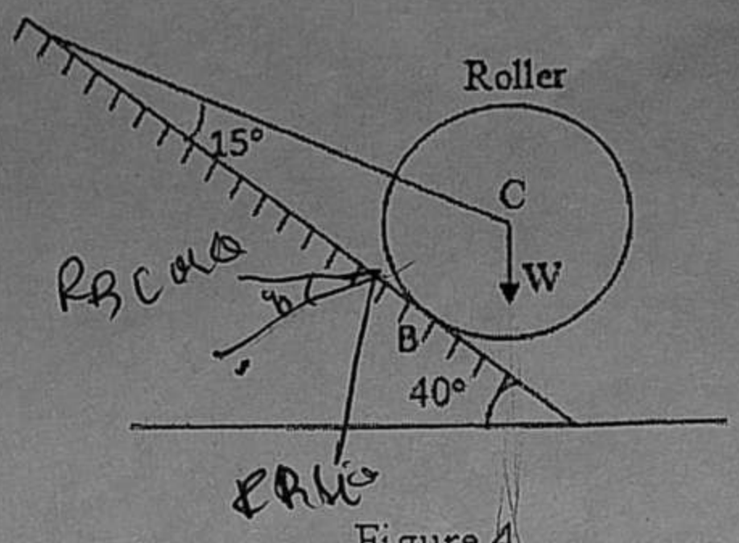


Figure 4

(c) A uniform ladder of mass 20 kg and length 5 m is placed against a smooth and frictionless vertical wall with its lower end 3.5 m away from the wall. If the ladder is just to slip, determine the coefficient of friction between the ladder and the floor.

Q3	(Attempt any Two Parts) UNIT-2	(5,5)

(a) A triangle has base width '*b*' and height '*h*'. Derive the expression for the distance of the centroid of the area of the triangle from its base.

(b) Determine the moment of Inertia I_{xx} and I_{yy} of the plane area shown in Figure 5 about its centroidal axes.

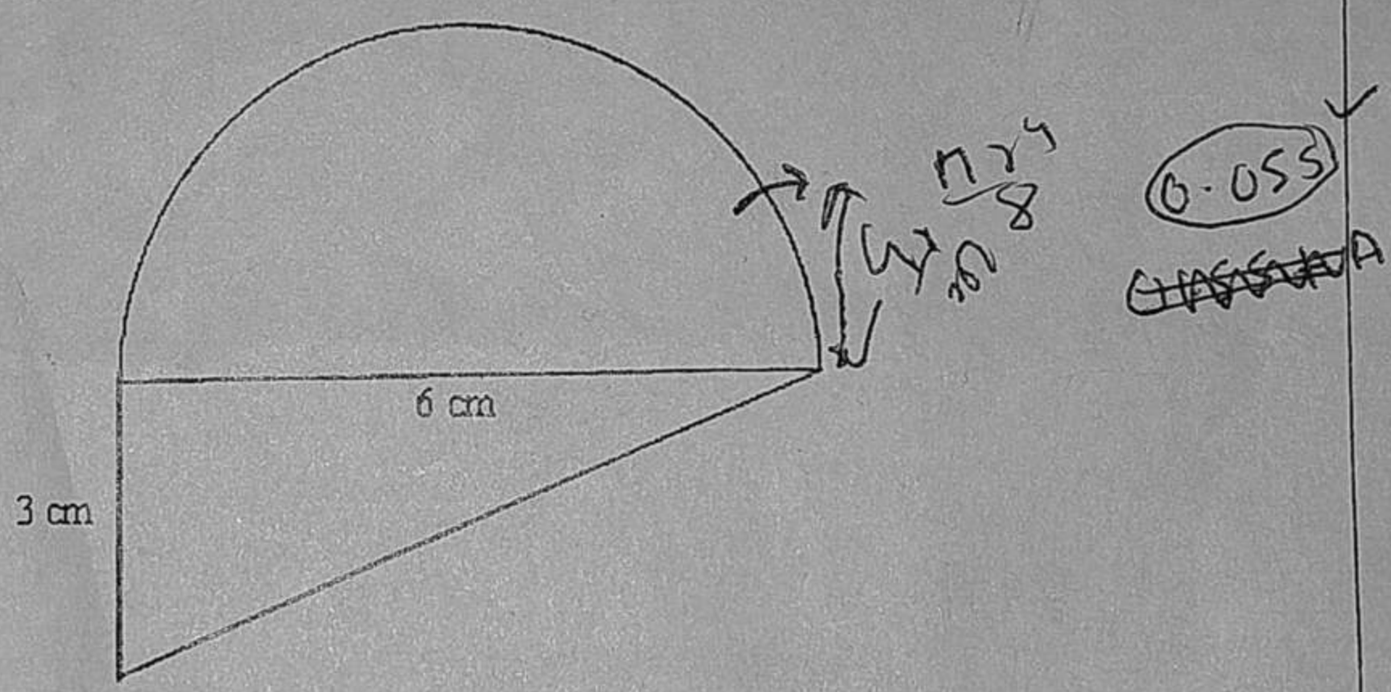


Figure 5

(c) Determine the forces in each member of the truss loaded as shown in Figure 6. Use method of joints.

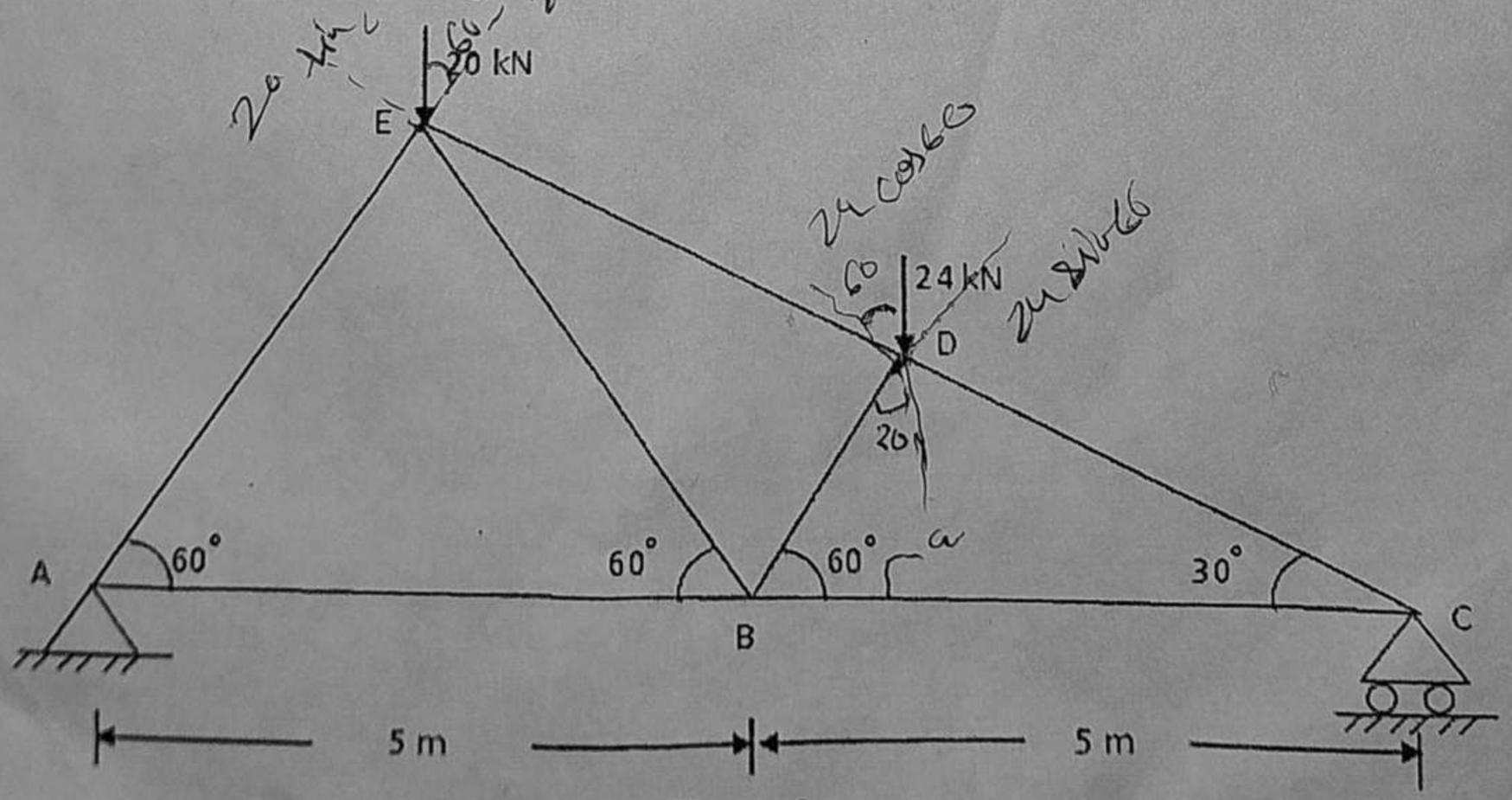


Figure 6